

Voronoi tessellations for protein interfaces and SEPIQ

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Life Sciences
Center

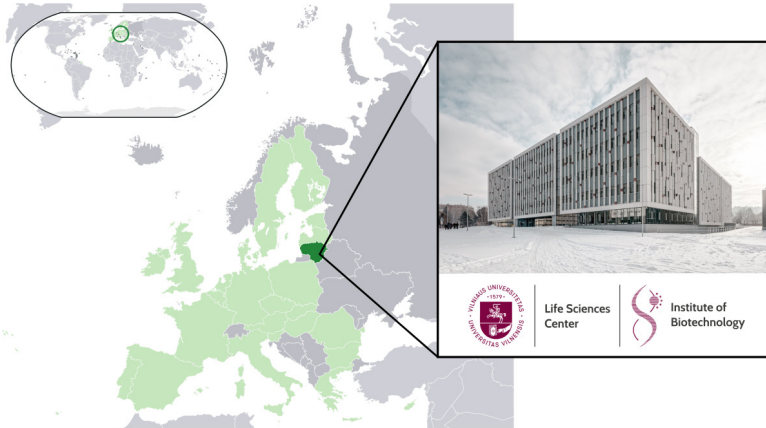


Institute of
Biotechnology

Where I am from

<https://gmc.vu.lt>

<https://bioinformatics.lt>

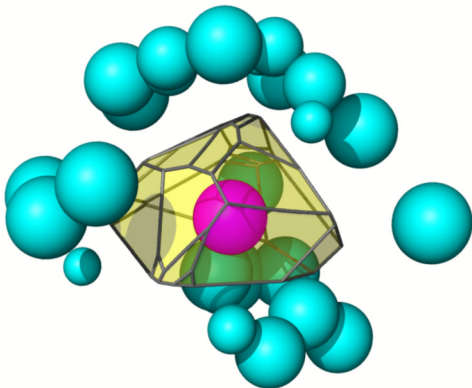


What I do

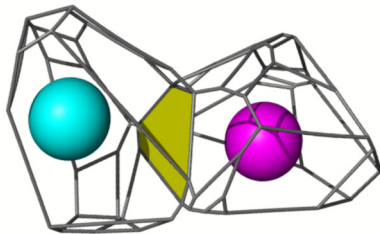
- ▶ I am a senior researcher in Vilnius University Life Sciences Center, I work on developing novel effective methods of **structural bioinformatics**.
- ▶ Most of my methods construct and utilize the **Voronoi tessellation** of atomic balls and the tessellation-derived interatomic contact areas.
- ▶ The developed tessellation-based methodology is especially efficient in analyzing **protein-protein interfaces**.
 - ▶ It was instrumental in successes of the Vilnius University team in recent community-wide CASP and CAPRI challenges (e.g. consistent top placements in CAPRI in 2016-2024).

Deriving atom-atom contact areas using the Voronoi tessellation

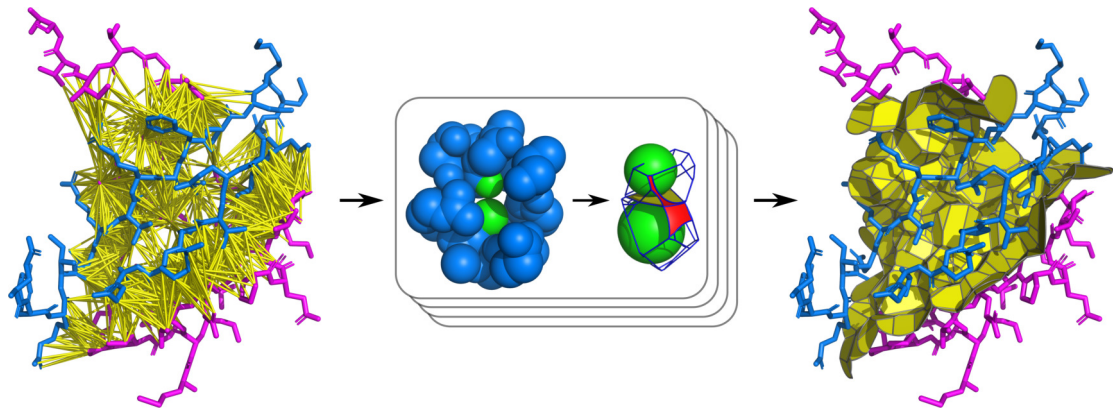
Voronoi cell of an atom surrounded by its neighbors



Atom-atom contact surface defined as the face shared by two adjacent Voronoi cells.



Deriving interface contacts using the Voronoi tessellation



Done using Voronota-LT. For more information about it and my research, please visit www.kliment.lt.

What I do with SEPIQ

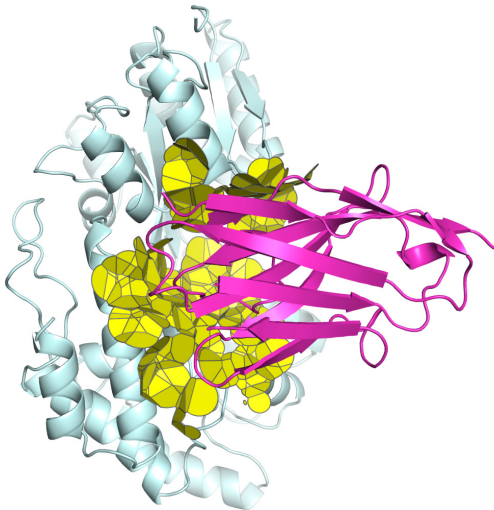
I am involved in the development of the framework for assessing predictions of antibody-antigen interface residues.

More, specifically, I consult on:

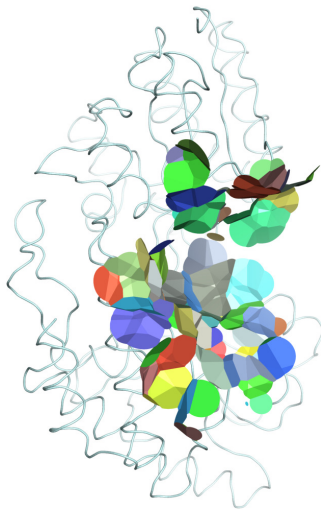
- ▶ how to describe the interface binding sites in the ground truth structure
- ▶ how to evaluate the prediction of the binding sites based on the ground truth description

SEPIQ-like example of structural interface ground truth

Inter-chain contact areas
in ground truth structure



The same contact areas
grouped per antigen residue



Thank you

Thank you SEPIQ team.

Thank you all for listening.